1. (a) $\exists$ a linear transformation $S: W \to V$. s.t.

$$T \circ S = id_W \quad S \circ T = id_V$$

where id_W and id_V are the identity transformations on W and V respectively.

Equivalently. T is invertible $\Leftrightarrow$ it is bijective.

(b)

$1 \Rightarrow 2$ Suppose T is invertible

Assume $B = \{b_1, \dots, b_n\}$. Then $C = \{T(b_1), \dots, T(b_n)\}$

① linear independence

Suppose $a_1 T(b_1) + \dots + a_n T(b_n) = 0$

$\Rightarrow T(a_1 b_1) + \dots + T(a_n b_n) = 0$.

$\Rightarrow T(a_1 b_1 + \dots + a_n b_n) = 0$.

Since T invertible $\Rightarrow T^{-1} T(a_1 b_1 + \dots + a_n b_n) = 0 \Rightarrow a_1 b_1 + \dots + a_n b_n = 0$

Since B linear independent $\Rightarrow a_1 = \dots = a_n = 0 \Rightarrow C$ linear independent

② spans W

Take $\forall w \in W. \Rightarrow \exists v \in V$ s.t. $T^{-1}(w) = v$. and $w = T(v)$

Since B basis, $\Rightarrow v = a_1 b_1 + \dots + a_n b_n = a_1 T^{-1}(T(b_1)) + \dots + a_n T^{-1}(T(b_n))$

$= T^{-1}(a_1 T(b_1) + \dots + a_n T(b_n))$.

$\Rightarrow T(v) = a_1 T(b_1) + \dots + a_n T(b_n)$

$\Rightarrow w = a_1 c_1 + \dots + a_n c_n$.

$\Rightarrow C$ spans W.

2 ⇒ 1

① Injective: Suppose $T(v)=0$. Then $T(a_1b_1+\cdots+a_nb_n)=0$.

$T(a_1b_1+\cdots+a_nb_n)=a_1T(b_1)+\cdots+a_nT(b_n)=0$ Since $C=\{T(b_1),\ldots,T(b_n)\}$ and C linear independent $\Rightarrow a_1=\cdots=a_n=0$.

$\Rightarrow \ker(T)=\{0\} \Rightarrow T$ injective.

② Surjective. Take $\forall w\in W$. Since $w=a_1T(b_1)+\cdots+a_nT(b_n)$

Then $w=T(a_1b_1+\cdots+a_nb_n)$ Since $B=\{b_1,\ldots,b_n\} \Rightarrow a_1b_1+\cdots+a_nb_n\in V$

$\Rightarrow T$ surjective.

Therefore T is invertible.

(6) 1.

Because $|S|=N$ and $\dim(H)=N$, it is suffice to prove S is linear independent.

Proof by Contradiction:

Let $a_0e_0+\cdots+a_{N-1}e_{N-1}=0$ WLOG assume $a_j\neq 0$.

$\Rightarrow a_je_j=-\sum_{i\neq j,\, i=0}^{N-1} a_ie_i$

$a_je_j\cdot\left(-\sum_{i=0,i\neq j}^{N-1} a_ie_i\right)=a_j^2\langle e_j,e_j\rangle=a_j^2>0$

However: $a_je_j\left(-\sum_{i=0,i\neq j}^{N-1} a_ie_i\right)=-\sum_{i=0,i\neq j}^{N-1} a_ia_je_ie_j=-\sum 0=0$.

Contradicts! $\Rightarrow a_j = 0$ for $\forall j$.

Therefore $\mathcal{E}$ is linear independent.

2.

1.

For $\forall f \in H$, we have $f = \sum_t f(t)\delta_t$.

Then $\hat{f} = \bar{F}_N[f] = \bar{F}_N(\sum_t f(t)\delta_t)$ Since $\bar{F}_N$ is linear transformation.

$$\Rightarrow \hat{f} = \bar{F}_N(f(0)\delta_0 + \cdots + f(N-1)\delta_{N-1}) = \sum_{t=0}^{N-1} f(t)\bar{F}_N(\delta_t) = \sum_{t=0}^{N-1} f(t)e_{-t}.$$

$$\Rightarrow \hat{f}(w) = \sum_{t=0}^{N-1} f(t)e_t(w) = \frac{1}{\sqrt{N}}\sum_{t=0}^{N-1} f(t)e^{-\frac{2\pi i}{N}wt}$$

2.

$f = \sum_{w\in Z_N} a_w e_w$, then we have.

$$\langle f, e_{w'}\rangle = \sum_{w\in Z_N} a_w \langle e_w, e_{w'}\rangle = a_{w'}$$

Therefore, $a_w = \langle f, e_w\rangle$

$$\Rightarrow f = \sum_{w\in Z_N} \langle f, e_w\rangle e_w \quad \text{Since } \langle f, e_w\rangle = \overline{\sum_{t\in Z_N} f\cdot \overline{e_w}}$$

$$= \frac{1}{\sqrt{N}}\sum_{t\in Z_N} f(t)e^{-\frac{2\pi i}{N}wt}$$

$$= \hat{f}(w).$$

$$\Rightarrow f = \sum_{w\in Z_N} \hat{f}(w)e_w.$$

3.

First we prove $\{\delta_t\}$ is basis of H.

① spanning: $\forall f \in H$. $f = \sum_{t=0}^{N-1} f(t)\,\delta_t$

② linear independence: $a_0\delta_0 + \cdots + a_{N-1}\delta_{N-1} = 0$

Take $i \in \mathbb{Z}_N$ $a_0\delta_0(i) + \cdots + a_{N-1}\delta_{N-1}(i) = 0 \Leftrightarrow a_i\delta_i(i) = a_i = 0$.

Then $i = 0, 1, 2, \cdots, N-1 \Rightarrow a_0 = a_1 = \cdots = a_{N-1} = 0$

$\Rightarrow \{\delta_0\}$ linear independent

Then we use 1(b). $\vec{F}_N$ maps $\{\delta_t\}$ of H to basis $\mathcal{E}$ of H.

$\Rightarrow F_N$ is invertible.

2.

(a) A scalar $\lambda \in \mathbb{F}$ is eigenvalue of T. if $\exists$ a vector $v \neq 0$ such that $Tv = \lambda v$

A vector v is called eigenvector with eigenvalue $\lambda \in \mathbb{F}$. if $v \neq 0$ and $Tv = \lambda v$

(b) Since B is a basis and consist of eigenvectors of T.

$\Rightarrow \forall i.\ Tv_i = \lambda_i v_i$

Because the ith column of $[T]_B$ is the coordinate vector of $T(v_i)$ relative to B.

If we use e_i is i^{th} standard basis vector in $\mathbb{F}^n$.

then $[T(v_i)]_B = [\lambda_i v_i]_B = \lambda_i e_i$

$\Rightarrow [T]_B = (\lambda_1 e_1, \cdots, \lambda_n e_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$

(c)

1. $L[f](t) = f(t-1)$

First we find the eigenvalues of L. $Lf = \lambda f \Rightarrow Lf(t) = \lambda f(t)$.

$\Rightarrow f(t-1) = \lambda f(t)$ for $\forall t$. $\Rightarrow L^2 f(t) = L[Lf](t) = Lg(t) = g(t-1) = Lf(t-1)$

$= f((t-1)-1) = f(t-2) \Rightarrow L^3 f(t) = f(t-3) = f(t)$ Since $H = C(\mathbb{R}^3)$

$\Rightarrow L^3 = I$.

$\Rightarrow L^3 f = f$

Since $L^3 f = L^2(Lf) = \lambda L(Lf) = \lambda^2 Lf = \lambda^3 f \Rightarrow \lambda^3 = 1$

$\Rightarrow \lambda_1 = 1, \quad \lambda_2 = e^{\frac{2}{3}\pi i} \quad \lambda_3 = e^{\frac{4}{3}\pi i}$

$$\Rightarrow [L]_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{\frac{2}{3}\pi i} & 0 \\ 0 & 0 & e^{\frac{4}{3}\pi i} \end{pmatrix}$$

Consider $f_0(t) = 1$ then $L[f_0](t) = f_0(t-1) = f_0(t)$

$f_1(t) = w^{-t}$ then $L[f_1](t) = f_1(t-1) = w f_1(t)$.

$f_2(t) = w^{-2t}$ then $L[f_2](t) = f_2(t-1) = w^2 f_2(t)$

$$f_0 = \begin{pmatrix} f_0(1) \\ f_0(2) \\ f_0(3) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad f_1 = \begin{pmatrix} f_1(0) \\ f_1(1) \\ f_1(2) \end{pmatrix} = \begin{pmatrix} 1 \\ w^2 \\ w \end{pmatrix} \quad f_2 = \begin{pmatrix} f_2(0) \\ f_2(1) \\ f_2(2) \end{pmatrix} = \begin{pmatrix} 1 \\ w \\ w^2 \end{pmatrix}$$

$$B = \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ w^2 \\ w \end{pmatrix} \begin{pmatrix} 1 \\ w \\ w^2 \end{pmatrix} \right) \quad [L]_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{\frac{2}{3}\pi i} & 0 \\ 0 & 0 & e^{\frac{4}{3}\pi i} \end{pmatrix}$$

2. Let $w = e^{\frac{2\pi i}{3}}$ $M[f](t) = w^t f(t)$.

$M[f] = \lambda f$ then $M[f](t) = \lambda f(t)$ for $\forall t \in \mathbb{Z}_3$.

$\Rightarrow w^t f(t) = \lambda f(t)$ for $\forall t \in \mathbb{Z}_3$

Assume $f = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ wb \\ w^2 c \end{pmatrix} = \begin{pmatrix} \lambda a \\ \lambda b \\ \lambda c \end{pmatrix}$

$$A_M f = A_M \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ wb \\ w^2 c \end{pmatrix} \Rightarrow A_M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & w & 0 \\ 0 & 0 & w^2 \end{pmatrix}$$

$$\Rightarrow A_M f = \lambda f \Rightarrow (1-\lambda)(w-\lambda)(w^2-\lambda) = 0 \Rightarrow \lambda_1 = 1,\ \lambda_2 = w,\ \lambda_3 = w^2$$

$$A_M f_1 = \begin{pmatrix} a \\ wb \\ w^2 c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow f_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad A_M f_2 = \begin{pmatrix} a \\ wb \\ w^2 c \end{pmatrix} = \begin{pmatrix} wa \\ wb \\ w^2 c \end{pmatrix} \Rightarrow f_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$A_M f_3 = \begin{pmatrix} a \\ wb \\ w^2 c \end{pmatrix} = \begin{pmatrix} w^2 a \\ w^2 b \\ w^2 c \end{pmatrix} \Rightarrow f_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \mathcal{B} = \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right)$$

$$[M]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & w & 0 \\ 0 & 0 & w^2 \end{pmatrix}$$

3.

(a) For every $\tau \in \mathbb{Z}_N$, we have an operator $L_\tau : H \to H$ called time-shift, given by

$$L_\tau[f](t) = f(t-\tau)$$

and for every $w \in \mathbb{Z}_N$, we have an operator $M_w : H \to H$ called frequency-shift, given by

$$M_w[f](t) = e^{\frac{2\pi i}{N} wt} f(t).$$

(b) Take $\forall f \in H$.

$$(M_w \circ L_\tau)[f](t) = e^{\frac{2\pi i}{N} wt} L_\tau[f](t) = e^{\frac{2\pi i}{N} wt} \cdot f(t-\tau)$$

$$e^{\frac{2\pi i}{N} w\tau} (L_\tau \circ M_w)[f](t) = e^{\frac{2\pi i}{N} w\tau} \cdot M_w[f](t-\tau) = e^{\frac{2\pi i}{N} w\tau} \cdot e^{\frac{2\pi i}{N} w(t-\tau)} f(t-\tau)$$

$$= e^{\frac{2\pi i}{N} wt} \cdot f(t-\tau)$$

$$\Rightarrow M_w \circ L_\tau = e^{\frac{2\pi i}{N} w\tau} \cdot L_\tau \circ M_w \quad \text{for every } \tau, w \in \mathbb{Z}_N.$$

(c) Take $\forall f \in H$. and $\forall t \in \mathbb{Z}_N$.

$$L_\tau \circ F_N[f](t) = F_N[f](t-\tau) = \frac{1}{\sqrt{N}} \sum_{s \in \mathbb{Z}_N} f(s) \cdot e^{-\frac{2\pi i}{N}(t-\tau)s}$$

$$= \frac{1}{\sqrt{N}} \sum_{s \in \mathbb{Z}_N} f(s) \cdot e^{\frac{2\pi i}{N} s(\tau - t)}$$

$$F_N \circ M_\tau[f](t) = \frac{1}{\sqrt{N}} \sum_{s \in \mathbb{Z}_N} M_\tau[f](s)\, e^{-\frac{2\pi i}{N} ts}$$

$$= \frac{1}{\sqrt{N}} \sum_{s \in \mathbb{Z}_N} e^{\frac{2\pi i}{N}\tau s} f(s)\, e^{-\frac{2\pi i}{N} ts}$$

$$= \frac{1}{\sqrt{N}} \sum_{s \in \mathbb{Z}_N} f(s)\, e^{\frac{2\pi i}{N} s(\tau - t)}$$

$$\Rightarrow L_\tau \circ F_N = F_N \circ M_\tau$$